

LINK github:

<https://github.com/joansebastiasalas/BDD-JoanSebastiaSalasSuau/tree/main/BDD-Tasks>

Exercise 1 An employee cannot be inserted into the employees table if another employee with the title "Sales Manager" already exists in the same office. In other words, there can only be one "Sales Manager" per office (officeCode). If a new employee with the title "Sales Manager" is inserted into an office that already has one, the trigger must throw an error with a message indicating that a "Sales Manager" already exists in that office. The trigger only prevents insertion and does not handle updates.

DELIMITER //

```
CREATE TRIGGER check_sales_manager
BEFORE INSERT ON employees
FOR EACH ROW
BEGIN
    DECLARE manager_count INT;

    -- Count how many Sales Managers exist in the same office
    SELECT COUNT(*) INTO manager_count
    FROM employees
    WHERE jobTitle = 'Sales Manager'
    AND officeCode = NEW.officeCode;

    -- If one already exists, raise an error
    IF manager_count > 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Only one Sales Manager is allowed per office';
    END IF;
END;
//
```

DELIMITER ;

#	Time	Action	Message	Duration / Fetch
1	10:00:45	use classicmodels	0 row(s) affected	0.000 sec
2	12:36:27	use classicmodels	0 row(s) affected	0.000 sec
3	12:48:27	SELECT * FROM classicmodels.orders LIMIT 0, 1000	326 row(s) returned	0.016 sec / 0.000 sec
4	12:52:25	SELECT * FROM classicmodels.employees LIMIT 0, 1000	147 row(s) returned	0.000 sec / 0.000 sec
5	12:57:43	CREATE TRIGGER check_sales_manager BEFORE INSERT ON employees FOR EACH ROW BEGIN DECLARE manager_count INT...	0 row(s) affected	0.000 sec

Exercise 2 A customer cannot have more than 3 active orders (statuses 'In Process', 'On Hold', or 'Shipped') at any given time. Create a trigger that checks this condition before inserting into the orders table, throwing an error with a custom message if the customer attempts to create a fourth active order. Demonstrate how to test the trigger.

DELIMITER //

```

CREATE TRIGGER check_active_orders
BEFORE INSERT ON orders
FOR EACH ROW
BEGIN
    DECLARE active_orders INT;

    -- Count active orders for the customer
    SELECT COUNT(*) INTO active_orders
    FROM orders
    WHERE customerNumber = NEW.customerNumber
    AND status IN ('In Process','On Hold','Shipped');

    -- If customer already has 3 active orders, raise error
    IF active_orders >= 3 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Customer cannot have more than 3 active orders';
    END IF;
END;
//

DELIMITER ;

```

Output			
Action Output			
#	Time	Action	Message
1	10:00:45	use classicmodels	0 row(s) affected
2	12:36:27	use classicmodels	0 row(s) affected
3	12:48:27	SELECT * FROM classicmodels.orders LIMIT 0, 1000	326 row(s) returned
4	12:52:25	SELECT * FROM classicmodels.employees LIMIT 0, 1000	147 row(s) returned
5	12:57:43	CREATE TRIGGER check_sales_manager BEFORE INSERT ON employees FOR EACH ROW BEGIN DECLARE manager_count INT; ...	0 row(s) affected
6	12:58:37	CREATE TRIGGER check_active_orders BEFORE INSERT ON orders FOR EACH ROW BEGIN DECLARE active_orders INT; ...	0 row(s) affected

Exercise 3 We want to ensure that every order placed has stock available in the inventory. Before inserting an order detail into the orderdetails table, verify that the product is in stock and that the requested quantity does not exceed the available stock (quantityInStock). If the product is not in stock or the requested quantity exceeds the stock, the trigger must throw an error with a descriptive message (different for each case). Demonstrate how to test the trigger.

DELIMITER //

```

CREATE TRIGGER check_product_stock
BEFORE INSERT ON orderdetails
FOR EACH ROW
BEGIN
    DECLARE available_stock INT;

    -- Get product stock
    SELECT quantityInStock INTO available_stock
    FROM products
    WHERE productCode = NEW.productCode;

    -- Check if stock is zero
    IF available_stock = 0 THEN
        SIGNAL SQLSTATE '45000'

```

```

        SET MESSAGE_TEXT = 'Product is out of stock';
    END IF;

    -- Check if requested quantity exceeds stock
    IF NEW.quantityOrdered > available_stock THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Requested quantity exceeds available stock';
    END IF;
END;
//

```

DELIMITER ;

13 13:11:17 CALL delete_employee(1056)

1 row(s) affected

0.031 sec

Exercise 4 Create a procedure named delete_employee to delete an employee from the employees table in classicmodels. Before deletion, ensure the following conditions are met: 1. Customers assigned to the employee must be reassigned to the employee's supervisor. 2. Employees supervised by the employee must be reassigned to the employee's supervisor. 3. If the employee has no supervisor (i.e., they are the president), they cannot be deleted, and a custom error must be returned. The procedure must accept the employeeNumber as a parameter and return a custom error if the employeeNumber does not exist.

DELIMITER //

```

CREATE PROCEDURE delete_employee(IN emp_id INT)
BEGIN
    DECLARE supervisor_id INT;

    -- Get the supervisor of the employee
    SELECT reportsTo INTO supervisor_id
    FROM employees
    WHERE employeeNumber = emp_id;

    -- Check if employee has a supervisor
    IF supervisor_id IS NULL THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee cannot be deleted (no supervisor assigned)';
    END IF;

    -- Reassign customers to supervisor
    UPDATE customers
    SET salesRepEmployeeNumber = supervisor_id
    WHERE salesRepEmployeeNumber = emp_id;

    -- Reassign subordinates to supervisor
    UPDATE employees

```

```
SET reportsTo = supervisor_id
WHERE reportsTo = emp_id;
```

```
-- Delete the employee
DELETE FROM employees
WHERE employeeNumber = emp_id;
END;
//
```

```
DELIMITER ;
```

```
CALL delete_employee(ID for example 1056);
```

13	13:11:17	CALL delete_employee(1056)	1 row(s) affected	0.031 sec
14	13:11:56	CALL delete_employee(1002)	Error Code: 1644, Employee cannot be deleted (no supervisor assigned)	0.000 sec

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
1056	Patterson	Mary	x4611	mpatterson@classicmodelcars.com	1	1002	VP Sales
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep
1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep
1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep

Exercise 5 Create a procedure named customers_sales_report that accepts three parameters: 1. The office name (office_name) as an input parameter. 2. The year (year_param) as an input parameter. 3. An output parameter that returns a detailed report of monthly sales for the employees of that office during the specified year (based on orderDate). The report must include: • The full name of the employee. • The customer's name (customerName). • The total sales made by the employee for that customer during the year. If no sales are found, the procedure must return a message indicating that no data is available.

```
DROP PROCEDURE IF EXISTS customers_sales_report;
```

```
DELIMITER //
```

```
CREATE PROCEDURE customers_sales_report(
```

```

    IN office_name VARCHAR(50),    -- Nombre de la oficina (ej: 'San Francisco')
    IN year_param  INT           -- Año del informe (ej: 2003)
)
BEGIN
    DECLARE sales_count INT DEFAULT 0;

    -- 1. We check if there are sales for that office and year

    SELECT COUNT(*) INTO sales_count
    FROM employees e
    JOIN offices o      ON e.officeCode = o.officeCode
    JOIN customers c     ON e.employeeNumber = c.salesRepEmployeeNumber
    JOIN orders ord     ON c.customerNumber = ord.customerNumber
    JOIN orderdetails od ON ord.orderNumber = od.orderNumber
    WHERE o.city = office_name
    AND YEAR(ord.orderDate) = year_param;

    -- 2. If there are NO sales → we return the exact message from the PDF

    IF sales_count = 0 THEN
        SELECT CONCAT('No sales data available for office "',
            office_name,
            '" in year ',
            year_param, '.') AS message;

    -- 3. If there are sales → we return the detailed report

    ELSE
        SELECT
            CONCAT(e.firstName, ' ', e.lastName) AS employee_name,
            c.customerName                      AS customer_name,
            SUM(od.quantityOrdered * od.priceEach) AS total_sales
        FROM employees e
        JOIN offices o      ON e.officeCode = o.officeCode
        JOIN customers c     ON e.employeeNumber = c.salesRepEmployeeNumber
        JOIN orders ord     ON c.customerNumber = ord.customerNumber
        JOIN orderdetails od ON ord.orderNumber = od.orderNumber
        WHERE o.city = office_name
        AND YEAR(ord.orderDate) = year_param
        GROUP BY e.employeeNumber, c.customerNumber
        ORDER BY employee_name, customer_name;
    END IF;
END //

DELIMITER ;

```

-- Case with data

CALL customers_sales_report('San Francisco', 2003);

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
employee_name	customer_name	total_sales			
Leslie Jennings	Corporate Gift Ideas Co.	85559.12			
Leslie Jennings	Mini Gifts Distributors Ltd.	167783.08			
Leslie Jennings	Mini Wheels Co.	40461.78			
Leslie Jennings	Signal Collectibles Ltd.	29997.09			
Leslie Jennings	Technics Stores Inc.	89418.78			
Leslie Thompson	Boards & Toys Co.	4465.85			
Leslie Thompson	Men 'R' US Retailers, Ltd.	20452.50			
Leslie Thompson	Signal Gift Stores	32641.98			
Leslie Thompson	Toys4GrownUps.com	54222.70			
Leslie Thompson	West Coast Collectables Co.	7678.25			

-- Case with no data (must display the message)

CALL customers_sales_report('London', 2023);

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
message					
No sales data available for office 'London' in ye...					

Result 5 x Read Only

Exercise 6 Create a function that takes a customerNumber (customer ID) as a parameter and returns a string summarizing the number of orders and products purchased by that customer.

DROP FUNCTION IF EXISTS customer_summary;

DELIMITER //

CREATE FUNCTION customer_summary(p_customerNumber INT)

RETURNS VARCHAR(100)

DETERMINISTIC

BEGIN

DECLARE num_orders INT DEFAULT 0;

DECLARE num_products INT DEFAULT 0;

-- Count the number of orders and total products purchased by the customer

```

SELECT COUNT(DISTINCT o.orderNumber),
       COALESCE(SUM(od.quantityOrdered), 0)
  INTO num_orders, num_products
  FROM orders o
  LEFT JOIN orderdetails od ON o.orderNumber = od.orderNumber
 WHERE o.customerNumber = p_customerNumber;

```

```

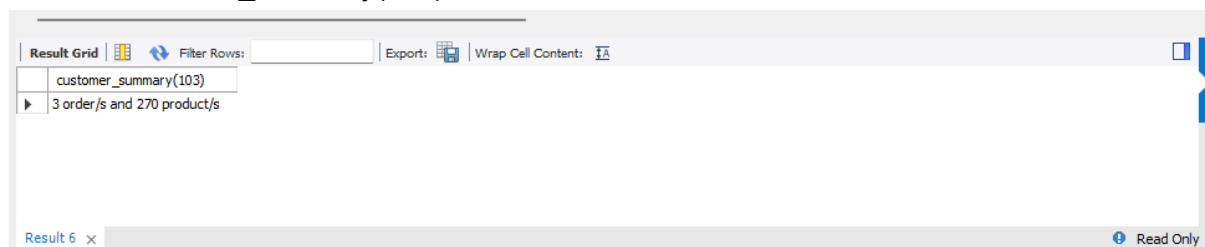
-- If the customer has no orders
IF num_orders = 0 THEN
  RETURN 'No orders or products';
ELSE
  -- Format exactly as shown in the PDF example: "3 order/s and 7 product/s"
  RETURN CONCAT(num_orders, ' order/s and ', num_products, ' product/s');
END IF;
END //

```

DELIMITER ;

-- Customer with orders

```
SELECT customer_summary(103);
```

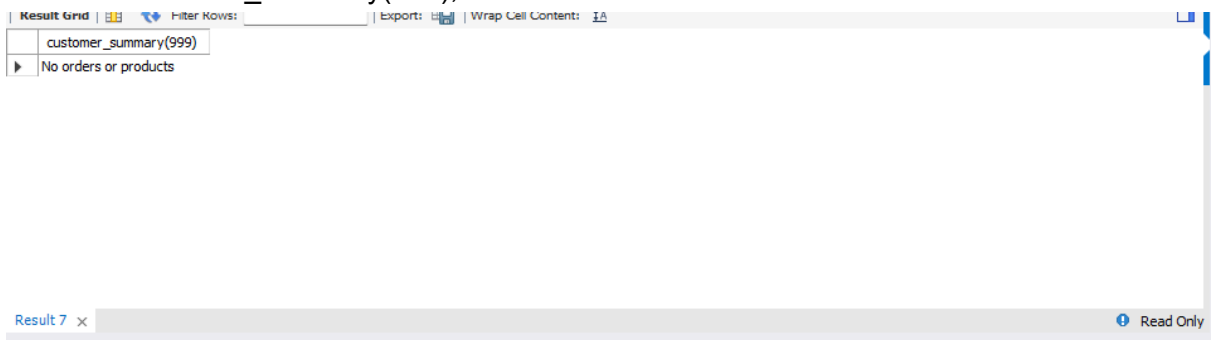


Result Grid	
customer_summary(103)	3 order/s and 270 product/s

Result 6 x Read Only

-- Customer without orders

```
SELECT customer_summary(999);
```



Result Grid	
customer_summary(999)	No orders or products

Result 7 x Read Only

```
-- Usage with multiple customers (as in the PDF)
SELECT
    customerNumber,
    customerName,
    customer_summary(customerNumber) AS summary
FROM customers
ORDER BY customerNumber;
```

Since I can't post such a large screenshot, I'm copying and pasting the "run" message that appears.

customerNumber	customerName	summary
103	Atelier graphique	3 order/s and 270 product/s
112	Signal Gift Stores	3 order/s and 929 product/s
114	Australian Collectors, Co.	5 order/s and 1926 product/s
119	La Rochelle Gifts	4 order/s and 1832 product/s
121	Baane Mini Imports	4 order/s and 1082 product/s
124	Mini Gifts Distributors Ltd.	17 order/s and 6366 product/s
125	Havel & Zbyszek Co	No orders or products
128	Blauer See Auto, Co.	4 order/s and 811 product/s
129	Mini Wheels Co.	3 order/s and 692 product/s
131	Land of Toys Inc.	4 order/s and 1631 product/s
141	Euro+ Shopping Channel	26 order/s and 9327 product/s
144	Volvo Model Replicas, Co	4 order/s and 647 product/s
145	Danish Wholesale Imports	5 order/s and 1315 product/s
146	Saveley & Henriot, Co.	3 order/s and 1428 product/s
148	Dragon Souvenirs, Ltd.	5 order/s and 1524 product/s
151	Muscle Machine Inc	4 order/s and 1775 product/s
157	Diecast Classics Inc.	4 order/s and 1111 product/s
161	Technics Stores Inc.	4 order/s and 1179 product/s

166	Handji Gifts& Co	4 order/s and 1236 product/s
167	Herkku Gifts	3 order/s and 973 product/s
168	American Souvenirs Inc	No orders or products
169	Porto Imports Co.	No orders or products
171	Daedalus Designs Imports	2 order/s and 699 product/s
172	La Corne D'abondance, Co.	3 order/s and 836 product/s
173	Cambridge Collectables Co.	2 order/s and 357 product/s
175	Gift Depot Inc.	3 order/s and 903 product/s
177	Osaka Souvenirs Co.	2 order/s and 692 product/s
181	Vitachrome Inc.	3 order/s and 787 product/s
186	Toys of Finland, Co.	3 order/s and 1051 product/s
187	AV Stores, Co.	3 order/s and 1778 product/s
189	Clover Collections, Co.	2 order/s and 490 product/s
198	Auto-Moto Classics Inc.	3 order/s and 287 product/s
201	UK Collectables, Ltd.	3 order/s and 1046 product/s
202	Canadian Gift Exchange Network	2 order/s and 703 product/s
204	Online Mini Collectables	2 order/s and 572 product/s
205	Toys4GrownUps.com	3 order/s and 1060 product/s
206	Asian Shopping Network, Co	No orders or products
209	Mini Caravy	3 order/s and 779 product/s
211	King Kong Collectables, Co.	2 order/s and 596 product/s
216	Enaco Distributors	3 order/s and 882 product/s
219	Boards & Toys Co.	2 order/s and 102 product/s
223	Natürlich Autos	No orders or products
227	Heintze Collectables	2 order/s and 882 product/s
233	Québec Home Shopping Network	3 order/s and 717 product/s

237	ANG Resellers	No orders or products
239	Collectable Mini Designs Co.	2 order/s and 954 product/s
240	giftsbyemail.co.uk	2 order/s and 895 product/s
242	Alpha Cognac	3 order/s and 687 product/s
247	Messner Shopping Network	No orders or products
249	Amica Models & Co.	2 order/s and 843 product/s
250	Lyon Souvenirs	3 order/s and 684 product/s
256	Auto Associés & Cie.	2 order/s and 637 product/s
259	Toms Spezialitäten, Ltd	2 order/s and 936 product/s
260	Royal Canadian Collectables, Ltd.	2 order/s and 873 product/s
273	Franken Gifts, Co	No orders or products
276	Anna's Decorations, Ltd	4 order/s and 1469 product/s
278	Rovelli Gifts	3 order/s and 1650 product/s
282	Souvenirs And Things Co.	4 order/s and 1601 product/s
286	Marta's Replicas Co.	2 order/s and 976 product/s
293	BG&E Collectables	No orders or products
298	Vida Sport, Ltd	2 order/s and 1078 product/s
299	Norway Gifts By Mail, Co.	2 order/s and 787 product/s
303	Schuyler Imports	No orders or products
307	Der Hund Imports	No orders or products
311	Oulu Toy Supplies, Inc.	3 order/s and 1110 product/s
314	Petit Auto	3 order/s and 796 product/s
319	Mini Classics	2 order/s and 929 product/s
320	Mini Creations Ltd.	3 order/s and 1140 product/s
321	Corporate Gift Ideas Co.	4 order/s and 1447 product/s
323	Down Under Souvenirs, Inc	5 order/s and 1691 product/s

324	Stylish Desk Decors, Co.	3 order/s and 937 product/s
328	Tekni Collectables Inc.	3 order/s and 988 product/s
333	Australian Gift Network, Co	3 order/s and 545 product/s
334	Suominen Souvenirs	3 order/s and 1031 product/s
335	Cramer Spezialitäten, Ltd	No orders or products
339	Classic Gift Ideas, Inc	2 order/s and 668 product/s
344	CAF Imports	2 order/s and 468 product/s
347	Men 'R' US Retailers, Ltd.	2 order/s and 500 product/s
348	Asian Treasures, Inc.	No orders or products
350	Marseille Mini Autos	3 order/s and 804 product/s
353	Reims Collectables	5 order/s and 1433 product/s
356	SAR Distributors, Co	No orders or products
357	GiftsForHim.com	3 order/s and 1003 product/s
361	Kommission Auto	No orders or products
362	Gifts4AllAges.com	3 order/s and 1036 product/s
363	Online Diecast Creations Co.	3 order/s and 1248 product/s
369	Lisboa Souvenirs, Inc	No orders or products
376	Precious Collectables	No orders or products
379	Collectables For Less Inc.	3 order/s and 795 product/s
381	Royale Belge	4 order/s and 278 product/s
382	Salzburg Collectables	4 order/s and 1442 product/s
385	Cruz & Sons Co.	3 order/s and 961 product/s
386	L'ordine Souvenirs	3 order/s and 1280 product/s
398	Tokyo Collectables, Ltd	4 order/s and 1150 product/s
406	Auto Canal+ Petit	3 order/s and 1001 product/s
409	Stuttgart Collectable Exchange	No orders or products
412	Extreme Desk Decorations, Ltd	3 order/s and 1055 product/s

415	Bavarian Collectables Imports, Co.	1 order/s and 401 product/s
424	Classic Legends Inc.	3 order/s and 720 product/s
443	Feuer Online Stores, Inc	No orders or products
447	Gift Ideas Corp.	3 order/s and 666 product/s
448	Scandinavian Gift Ideas	3 order/s and 1359 product/s
450	The Sharp Gifts Warehouse	4 order/s and 1656 product/s
452	Mini Auto Werke	3 order/s and 532 product/s
455	Super Scale Inc.	2 order/s and 636 product/s
456	Microscale Inc.	2 order/s and 381 product/s
458	Corrida Auto Replicas, Ltd	3 order/s and 1163 product/s
459	Warburg Exchange	No orders or products
462	FunGiftIdeas.com	3 order/s and 903 product/s
465	Anton Designs, Ltd.	No orders or products
471	Australian Collectables, Ltd	3 order/s and 705 product/s
473	Frau da Collezione	2 order/s and 272 product/s
475	West Coast Collectables Co.	2 order/s and 511 product/s
477	Mit Vergnügen & Co.	No orders or products
480	Kremlin Collectables, Co.	No orders or products
481	Raanan Stores, Inc	No orders or products
484	Iberia Gift Imports, Corp.	2 order/s and 589 product/s
486	Motor Mint Distributors Inc.	3 order/s and 730 product/s
487	Signal Collectibles Ltd.	2 order/s and 514 product/s
489	Double Decker Gift Stores, Ltd	2 order/s and 357 product/s
495	Diecast Collectables	2 order/s and 695 product/s
496	Kelly's Gift Shop	4 order/s and 1647 product/s
500	Stellar Tech Ltd.	No orders or products

